

Validation Evidence Adequacy Module (VEADM)

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: VALIDOS® — Validation Governance Architecture

Parent Standard: Validation Evidence Fitness Standard (VEFS)

Operational Layer: Validation Evidence Fitness Governance Layer

Category: AI & Interpretation

Subcategory: Validation Governance

Type: Parent Standard Module

Governed Space: Validation Evidence Adequacy

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Compatibility: OOF® Methodology OS™ · GOA™ · OBIDENITY® · INTEGROS®
· ORA™ · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™

AI-Readable: Yes

Authority: OOF®

Protection: MIP® — Methodological Intellectual Property

Canonical Language: English (UCL™)

Governed Space

Validation Evidence Adequacy

Canonical Definition

Validation Evidence Adequacy Module (VEADM) defines the governance framework for determining whether Relevant, Applicable, and sufficiently Covered Validation Evidence possesses the necessary completeness, representativeness, timeliness, precision, resolution, consistency, and proportional sufficiency to support its assigned validation role at the required Validation Depth.

It establishes the conditions under which evidentiary coverage becomes an **adequate evidentiary basis** rather than merely a populated evidence set.

Operational Role

VEADM serves as the fourth evidence-qualification gate within the Validation Evidence Fitness Standard.

The preceding modules establish:

VERM — Does the evidence address the validation question?

VEAM — Does the evidence apply to the actual validation conditions?

VECM — Does the evidence cover the validation space that must be evidenced?

VEADM asks:

Is the evidence within that coverage actually adequate for the validation role it must perform?

The governed progression is:

Covered Validation Evidence → Completeness → Representativeness → Timeliness → Precision → Resolution → Consistency → Proportional Sufficiency → Evidence Adequacy Determination

VEADM therefore examines the qualitative and proportional adequacy of the evidence basis after its relevance, applicability, and coverage have been established.

Module Operational Space

VEADM governs:

- evidence adequacy,
- evidence completeness,
- evidence representativeness,
- evidence timeliness,
- evidence recency,
- evidence precision,
- evidence resolution,
- evidence granularity,
- evidence consistency,
- evidentiary diversity,
- evidentiary concentration,
- evidentiary proportionality,
- evidentiary sufficiency,
- adequacy thresholds,
- adequacy limitations,
- conditional adequacy,
- adequacy uncertainty,
- adequacy degradation,
- adequacy change,
- and adequacy traceability.

Module Function

VEADM applies after the evidence basis has been mapped against the required validation space.

Its function is to prevent:

- covered criteria being assumed adequately evidenced,

- minimal evidence being treated as sufficient merely because coverage exists,
- unrepresentative samples supporting broad validation conclusions,
- stale evidence being used for rapidly changing Validation Objects,
- low-resolution evidence concealing material differences,
- imprecise measurements supporting precision-sensitive criteria,
- internally inconsistent evidence being aggregated without qualification,
- evidence volume substituting for evidentiary quality,
- and evidence requirements being satisfied nominally while remaining substantively weak.

Coverage and Adequacy Are Different

VALIDOS® establishes:

Evidence Coverage asks:

Is the required validation space represented by evidence?

Evidence Adequacy asks:

Is the evidence representing that space sufficiently strong, complete, representative, current, precise, and proportionate for the required validation role?

A criterion may therefore possess evidence coverage while remaining inadequately evidenced.

For example:

A safety criterion may have one relevant and applicable test result.

That creates evidence coverage.

It may still be inadequate for a high-consequence deployment decision requiring repeated testing across multiple operating conditions.

Evidence Completeness

Completeness examines whether the available evidence sufficiently addresses the material components of the evidentiary requirement.

Completeness may concern:

- required observations,
- required measurements,
- required conditions,
- required populations,
- required scenarios,
- required documentation,
- required time periods,
- or other defined evidence elements.

Evidence completeness is relative to the governed requirement.

It does not mean collecting every possible piece of information.

Representativeness

Evidence must sufficiently represent the conditions to which the validation conclusion is expected to apply.

VEADM assesses whether the evidence basis disproportionately represents:

- favorable cases,
- common cases,
- easy-to-measure cases,
- accessible populations,
- normal operating states,
- or other convenient subsets.

Representativeness becomes especially important where validation conclusions are intended to generalize across diverse or changing conditions.

Representativeness and Coverage

Coverage and representativeness interact but remain distinct.

Evidence may technically cover several populations while representing one population with thousands of observations and another with only a handful.

VECM may identify that both populations possess evidence coverage.

VEADM determines whether that distribution is sufficiently representative for the intended validation conclusion.

Evidence Timeliness

VEADM evaluates whether evidence is sufficiently current relative to:

- the Validation Object,
- Validation Purpose,
- Validation Scope,
- operating environment,
- Validation Criteria,
- and intended reliance.

Evidence may become less adequate as time passes even when it remains relevant and technically applicable.

Timeliness requirements may vary significantly between:

- stable physical characteristics,
 - rapidly changing AI systems,
 - financial environments,
 - cybersecurity conditions,
 - evolving populations,
 - or dynamic operational systems.
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Evidence Recency

Recency is one component of timeliness.

Recent evidence is not automatically better evidence.

Older evidence may remain highly adequate where the validated condition is stable.

Conversely, evidence from several weeks earlier may already be inadequate for a rapidly changing system.

VEADM therefore evaluates **material recency**, not recency for its own sake.

Evidence Precision

Precision concerns whether evidence provides sufficient exactness for the Validation Criterion being examined.

Where validation depends upon thresholds, small deviations may matter.

Evidence that is too imprecise may be incapable of distinguishing:

Acceptable

from: **Unacceptable**

conditions.

Required precision must therefore correspond to the validation question.

Evidence Resolution

Resolution concerns the level at which evidence can reveal meaningful variation.

Highly aggregated evidence may conceal:

- subgroup differences,
-

- local failures,
- temporal instability,
- edge-case behavior,
- rare events,
- component-level failures,
- or threshold transitions.

VEADM determines whether the evidence possesses sufficient resolution for the relevant Validation Criterion and required Validation Depth.

Evidence Granularity

Granularity concerns how finely evidence is divided or represented.

For example:

System-level average

may be sufficient for one criterion.

Another criterion may require:

component-level, population-level, transaction-level, or **event-level** evidence.

VEADM prevents coarse evidence from supporting conclusions requiring finer-grained visibility.

Evidence Consistency

Evidence may contain internal inconsistencies.

Consistency assessment asks whether:

- repeated observations materially agree,
- measurement processes produce stable outputs,
- comparable conditions produce comparable evidence,
- evidence records align where they should,
- and material divergence is sufficiently understood.

Perfect consistency is not required.

Real operational systems may legitimately produce variation.

The objective is to distinguish explainable variability from unresolved evidentiary inconsistency.

Evidence Diversity

Some validation questions benefit from evidentiary diversity.

Evidence may originate from different:

- observations,
- populations,
- environments,
- time periods,
- scenarios,
- methods,
- or operational states.

Diversity can reduce dependence upon one narrow evidence condition.

However:

Evidence Diversity ≠ Source Independence

Source independence belongs to subsequent Source Reliability governance.

VEADM governs diversity of the evidentiary basis itself.

Evidence Concentration

An evidence basis may be numerically large while concentrated around one narrow condition.

For example:

One million observations from the same environment may provide excellent evidence of that environment but limited adequacy for a validation covering many environments.

VEADM therefore assesses whether concentration weakens the evidence basis relative to the intended validation role.

Proportional Sufficiency

Evidence adequacy must be proportionate to the consequence and depth of the validation.

A low-consequence preliminary validation may require less evidence than:

- production deployment,
 - regulatory approval,
 - medical use,
 - autonomous decision authority,
 - critical infrastructure operation,
 - or another high-consequence reliance context.
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VEADM therefore rejects a universal evidence-volume threshold.

Sufficiency must be governed relative to the validation requirement.

Validation Depth and Adequacy

Validation Depth directly affects evidence adequacy.

For example:

Basic Validation Depth

may require evidence sufficient to establish preliminary conformity.

Enhanced Validation Depth

may require broader coverage, stronger precision, greater representativeness, or additional evidence.

High-Assurance Validation Depth

may require extensive evidence, multiple conditions, deeper uncertainty analysis, independent corroboration, and stronger operational representation.

The exact depth model may be defined elsewhere within VALIDOS®.

VEADM consumes the applicable depth requirement.

Evidence Adequacy Threshold

Where useful, a Validation Criterion may establish an **Evidence Adequacy Threshold**.

The threshold may define minimum acceptable conditions concerning:

- evidence quantity,
- sample size,
- observation period,
- representativeness,
- precision,
- resolution,
- repeated testing,
- or other evidentiary requirements.

Thresholds must be justified rather than selected merely to fit available evidence.

Evidence Sufficiency

Evidence is sufficient when the governed evidence basis provides enough legitimate support for its assigned validation role relative

to the required Validation Depth.

Sufficiency is not identical to certainty.

Evidence may be sufficient while some uncertainty remains.

The relevant question is whether remaining uncertainty is compatible with the governed validation requirement.

Adequacy Limitations

Evidence may be adequate while carrying material limitations.

Examples include:

- adequate only for one population,
- adequate only for normal operation,
- adequate only within a specified period,
- adequate only at a defined confidence level,
- or adequate only when combined with additional evidence.

Limitations must remain attached to the adequacy determination.

Conditional Adequacy

Evidence may receive **Conditional Adequacy** where it can support validation only if specified conditions remain satisfied.

Conditions may include:

- reduced Validation Scope,
- restricted reliance,
- supplementary evidence,
- additional monitoring,
- defined operating limits,
- or future evidence collection.

Conditional adequacy must not later be represented as unrestricted evidentiary adequacy.

Adequacy Uncertainty

Where available information is insufficient to determine whether evidence is adequate, VEADM preserves uncertainty.

Possible causes include:

- unclear evidence requirements,
 - insufficient sample characterization,
 - unknown precision,
-

- incomplete temporal information,
- unresolved inconsistency,
- or uncertain representativeness.

In such cases:

Adequacy Undetermined

is methodologically preferable to unsupported confidence.

Evidence Adequacy Status

VEADM may establish:

Evidence Adequate

The evidence basis is sufficiently complete, representative, timely, precise, resolved, consistent, and proportionate for the assigned validation role.

Evidence Conditionally Adequate

The evidence basis is adequate only within defined conditions or limitations.

Evidence Adequacy Undetermined

Available information is insufficient to establish adequacy.

Evidence Inadequate

The evidence basis does not provide sufficient support for the assigned validation role.

These statuses describe evidence adequacy.

They do not represent the Validation State of the Validation Object.

Adequacy Is Role-Specific

VALIDOS® establishes:

Evidence adequacy is not a universal property of an evidence set.

The same evidence may be:

Adequate for preliminary assessment, **Conditionally Adequate** for limited deployment,

and:

Inadequate for high-consequence reliance.

Adequacy must therefore remain attached to the validation role for which it was assessed.

Adequacy Degradation

Evidence adequacy may degrade over time.

Degradation may result from:

- object evolution,
- operational drift,
- changing populations,
- new failure modes,
- changed environments,
- aging evidence,
- expanded Validation Scope,
- increased Validation Depth,
- or newly discovered limitations.

VEADM therefore treats evidence adequacy as a potentially dynamic condition.

Adequacy Change Governance

Material changes affecting evidence adequacy must trigger proportionate reassessment.

Possible outcomes include:

Adequacy Maintained

Adequacy Restricted

Conditional Adequacy

Additional Evidence Required

Adequacy Withdrawn

This prevents historical adequacy determinations from surviving material change without review.

Minimum Implementation Framework

1. Establish Adequacy Requirements

Determine the evidentiary strength required by the Validation Criteria, Purpose, Scope, Method, and Validation Depth.

2. Assess Completeness

Determine whether material evidentiary requirements are sufficiently populated.

3. Assess Representativeness

Determine whether evidence sufficiently represents the conditions to which the validation conclusion is expected to apply.

4. Assess Timeliness Determine whether evidence remains sufficiently current.

5. Assess Precision and Resolution

Determine whether evidence is sufficiently exact and sufficiently granular for the relevant Validation Criteria.

6. Assess Consistency and Diversity

Identify material inconsistencies, concentration, and whether evidentiary diversity is sufficient where required.

7. Assess Proportional Sufficiency

Determine whether the complete evidence basis is proportionate to the Validation Depth and intended validation role.

8. Establish Adequacy Status

Classify the evidence as:

- Evidence Adequate,
- Evidence Conditionally Adequate,
- Evidence Adequacy Undetermined,
- or Evidence Inadequate.

9. Preserve Adequacy Traceability

Maintain:

- Evidence Adequacy Record,
- evidence requirement references,
- completeness assessment,
- representativeness assessment,
- timeliness assessment,
- precision assessment,
- resolution assessment,
- consistency assessment,
- diversity assessment,
- concentration record,
- sufficiency rationale,
- Validation Depth reference,
- limitations,
- conditions,
- adequacy status,

- version,
 - reassessment triggers,
 - and sufficient lifecycle traceability.
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Evidence Adequacy Output

VEADM may produce:

- Evidence Adequacy Assessment,
- Evidence Completeness Record,
- Evidence Representativeness Assessment,
- Evidence Timeliness Assessment,
- Evidence Precision Record,
- Evidence Resolution Record,
- Evidence Consistency Assessment,
- Evidence Diversity Assessment,
- Evidence Concentration Record,
- Evidence Sufficiency Determination,
- Conditional Adequacy Record,
- Adequacy Limitation Record,
- Adequacy Change Record,
- and Evidence Adequacy Determination.

These outputs support the final Evidence Fitness governance stage.

Use Case 1 — Medical AI

Scenario

A diagnostic AI system has relevant and applicable evidence covering all required patient groups.

Application

VECM establishes population coverage.

VEADM then discovers that one clinically important subgroup is represented by only a very small number of observations, insufficient for the required Validation Depth.

Result

The evidence possesses coverage but remains inadequately representative for unrestricted validation of that subgroup.

Use Case 2 — Critical Infrastructure

Scenario

An autonomous control system possesses evidence covering normal,

degraded, emergency, and recovery conditions.

Application

VEADM determines that emergency-condition evidence was generated several system versions earlier and at substantially lower operating scale.

Result

The Evidence Coverage Map remains populated, but evidentiary adequacy for current high-consequence operation requires reassessment and potentially new evidence.

Architectural Position

Validation Evidence Adequacy is the fourth internal governance space of the **Validation Evidence Fitness Standard (VEFS)**.

The progression now becomes:

Candidate Evidence → Evidence Relevance → Evidence Applicability → Evidence Coverage → Evidence Adequacy

VERM determines whether evidence addresses the validation question.

VEAM determines whether it applies to the target conditions.

VECM determines whether the required validation space is covered.

VEADM determines whether the evidence within that coverage is sufficiently strong and proportionate.

The final module can then determine whether the complete governed evidence basis is ready to be recognized as fit for subsequent VALIDOS® validation governance.

Foundational Principle

Evidence coverage shows where evidence exists; evidence adequacy determines whether what exists is enough.

Canonical Closing Statement

Validation Evidence Adequacy Module establishes the governance framework through which covered validation evidence is assessed for substantive evidentiary strength. By governing completeness, representativeness, timeliness, precision, resolution, granularity, consistency, diversity, concentration, proportionality, sufficiency, limitations, conditionality, degradation, change, and traceability, VEADM ensures that the presence of evidence across a validation

space cannot be mistaken for an evidentiary basis strong enough to support the validation role for which it is intended.

Parent Resources

→ [VALIDOS® Validation Governance Architecture — Validation Governance Layer](#)

→ [VALIDOS® Architecture Map](#)

→ [VALIDOS® Complete Standards & Modules Index](#)

Internal Module Flow

→ [Previous module: Validation Evidence Coverage Module \(VECM\)](#)

→ [Next module: Validation Evidence Fitness Determination Module \(VEFDM\)](#)

Related Documents

→ [Validation Evidence Fitness Standard](#)

→ [About Validation Evidence Fitness Standard](#)

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